

2.6 Moment generating functions and independence

Means and covariance matrices summarize a distribution. They do not determine it, and they cannot certify independence. For that we need a transform of the whole distribution. This section collects the facts that [Chapter 3](#) and [Chapter 4](#) use.

Definition 2.6.1 — Moment generating function

The **moment generating function** of a $p \times 1$ random vector $\mathbf{Y}$ is

$$M_{\mathbf{Y}}(\mathbf{t}) = \mathbf{E} \exp(\mathbf{t}^T \mathbf{Y}), \quad \mathbf{t} \in \mathbb{R}^p,$$

where the expectation may be $+\infty$. We say that $M_{\mathbf{Y}}$ *exists* if it is finite for all $\mathbf{t}$ in some neighbourhood $\{\|\mathbf{t}\| < h\}$ of the origin, $h > 0$.

Theorem 2.6.1 — Properties of moment generating functions

Suppose $M_{\mathbf{Y}}$ exists.

- (a) *Uniqueness.* If $M_{\mathbf{V}}$ also exists and $M_{\mathbf{Y}} = M_{\mathbf{V}}$ on a neighbourhood of $\mathbf{0}$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.
- (b) *Moments.* All moments of $\mathbf{Y}$ are finite. The gradient of $M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\mathbf{E}(\mathbf{Y})$, and its Hessian at $\mathbf{0}$ is $\mathbf{E}(\mathbf{Y}\mathbf{Y}^T)$. The Hessian of $\log M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\text{Cov}(\mathbf{Y})$.
- (c) *Affine maps.* For a constant $k \times p$ matrix $\mathbf{A}$ and $k \times 1$ vector $\mathbf{b}$, $M_{\mathbf{A}\mathbf{Y}+\mathbf{b}}(\mathbf{s}) = \exp(\mathbf{s}^T \mathbf{b}) M_{\mathbf{Y}}(\mathbf{A}^T \mathbf{s})$.
- (d) *Independence.* Partition $\mathbf{Y} = (\mathbf{Y}_1^T, \mathbf{Y}_2^T)^T$ and $\mathbf{t} = (t_1^T, t_2^T)^T$ conformably. Then $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(t_1) M_{\mathbf{Y}_2}(t_2)$ for all $\mathbf{t}$ in a neighbourhood of $\mathbf{0}$.

Proof. For (a), the scalar case is a theorem of analysis: a moment generating function that is finite near the origin extends analytically, which pins down the characteristic function and hence the distribution (Billingsley 1995). For vectors, fix $\mathbf{a}$. For small $|s|$, $M_{\mathbf{a}^T \mathbf{Y}}(s) = M_{\mathbf{Y}}(s\mathbf{a}) = M_{\mathbf{V}}(s\mathbf{a}) = M_{\mathbf{a}^T \mathbf{V}}(s)$, so $\mathbf{a}^T \mathbf{Y}$ and $\mathbf{a}^T \mathbf{V}$ have the same distribution by the scalar case, and [Proposition 2.6.2](#) below finishes the argument. For (b), finiteness near $\mathbf{0}$ implies $\mathbf{E} \exp(h'|Y_i|) < \infty$ for some $h' > 0$ and each i , so every polynomial in the components is integrable and differentiation under the expectation is justified. Then $\partial M / \partial t_i = \mathbf{E}(Y_i e^{t^T \mathbf{Y}})$ and $\partial^2 M / \partial t_i \partial t_j = \mathbf{E}(Y_i Y_j e^{t^T \mathbf{Y}})$. At $\mathbf{t} = \mathbf{0}$ these are $\mathbf{E}(Y_i)$ and $\mathbf{E}(Y_i Y_j)$. For $K = \log M$, the chain rule gives $\nabla^2 K = M^{-1} \nabla^2 M - M^{-2} (\nabla M)(\nabla M)^T$, which at $\mathbf{0}$, where $M = 1$, is $\mathbf{E}(\mathbf{Y}\mathbf{Y}^T) - \boldsymbol{\mu}\boldsymbol{\mu}^T = \text{Cov}(\mathbf{Y})$. For (c), $\mathbf{E} \exp(\mathbf{s}^T (\mathbf{A}\mathbf{Y} + \mathbf{b})) = e^{\mathbf{s}^T \mathbf{b}} \mathbf{E} \exp((\mathbf{A}^T \mathbf{s})^T \mathbf{Y})$. For (d), independence gives $\mathbf{E}(e^{t_1^T \mathbf{Y}_1} e^{t_2^T \mathbf{Y}_2}) = \mathbf{E}(e^{t_1^T \mathbf{Y}_1}) \mathbf{E}(e^{t_2^T \mathbf{Y}_2})$. Conversely, let $\mathbf{V}_1$ and $\mathbf{V}_2$ be independent with the distributions of $\mathbf{Y}_1$ and $\mathbf{Y}_2$. The vector $(\mathbf{V}_1^T, \mathbf{V}_2^T)^T$ has moment generating function $M_{\mathbf{Y}_1}(t_1) M_{\mathbf{Y}_2}(t_2)$, which by hypothesis equals $M_{\mathbf{Y}}$ near $\mathbf{0}$. By (a) it has the same joint distribution as $\mathbf{Y}$, so $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent. ■

The next fact explains why [Chapter 3](#) can define the multivariate normal through linear combinations.

Proposition 2.6.2 — Distributions are determined by linear combinations

If $\mathbf{a}^T \mathbf{Y}$ and $\mathbf{a}^T \mathbf{V}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^p$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.

Proof. The characteristic function $\phi_{\mathbf{Y}}(\mathbf{t}) = \mathbf{E} \exp(i\mathbf{t}^T \mathbf{Y})$ exists for every random vector and determines its distribution (Billingsley 1995). Since $\phi_{\mathbf{Y}}(\mathbf{t}) = \phi_{\mathbf{t}^T \mathbf{Y}}(1) = \phi_{\mathbf{t}^T \mathbf{V}}(1) = \phi_{\mathbf{V}}(\mathbf{t})$ for every $\mathbf{t}$, the two distributions coincide. ■

If $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent, so are $g(\mathbf{Y}_1)$ and $h(\mathbf{Y}_2)$ for measurable g and h . Independence of statistics is usually proved this way. For normal vectors, Theorem 3.4.1 reduces the independence of two blocks to a zero cross-covariance, and Theorem 4.4.1 and Theorem 4.4.2 build on that.

Independence of *several* vectors means that the joint distribution factors into all the marginals, not merely that each pair is independent.

Example. (Pairwise but not mutually independent)

Let S_1 and S_2 be independent, each equal to ± 1 with probability $\frac{1}{2}$, and put $S_3 = S_1 S_2$. Each S_j is ± 1 with probability $\frac{1}{2}$. Each pair is independent: for instance, $\Pr(S_1 = s, S_3 = u) = \Pr(S_1 = s, S_2 = su) = \frac{1}{4}$ for all signs s, u . So $\text{Cov}(\mathbf{S}) = \mathbf{I}_3$. But the three are not mutually independent, since S_3 is a function of the other two. A third-order moment shows the failure directly: $\mathbf{E}(S_1 S_2 S_3) = 1$, whereas mutual independence would force it to be 0. No condition on the covariance matrix can detect this, because covariances involve only pairs.

2.6.1 Exercises

2.6.1.1 A. Check your understanding

Exercise A1.

Find the moment generating function of the multinomial count vector of Example (Multinomial counts), and recover its mean and covariance matrix from [Theorem 2.6.1\(b\)](#).

2.6.1.2 B. Practice

Exercise B1.

Let $S_1, \dots, S_k$ be independent random signs. For each nonempty $A \subseteq \{1, \dots, k\}$ put $T_A = \prod_{i \in A} S_i$. Show that the $2^k - 1$ variables T_A are pairwise independent with covariance matrix $\mathbf{I}$, and find a triple of them that is not mutually independent.